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應用於醫學影像分割的先驗調制擴散模型
Modulated Prior Diffusion for Medical Image Segmentation

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中華民國 一一五年三月
March 2026

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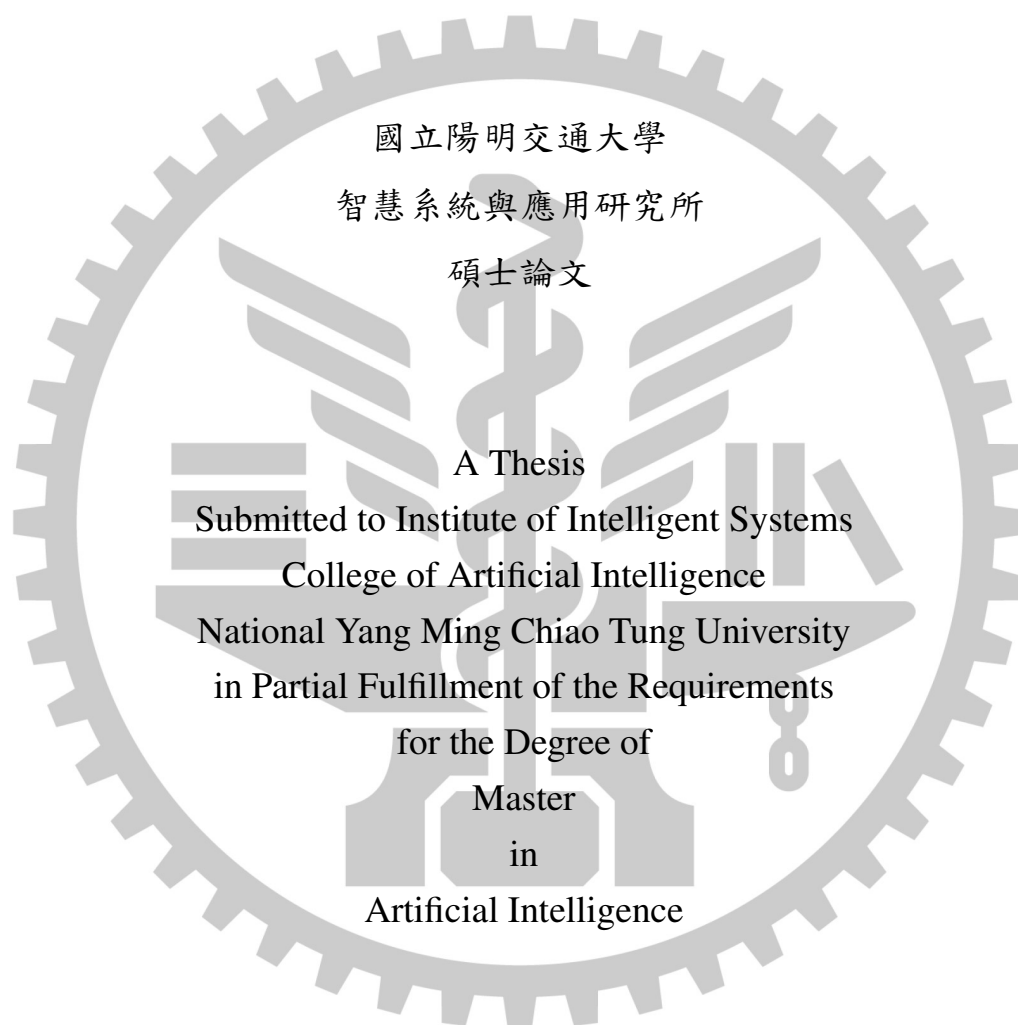
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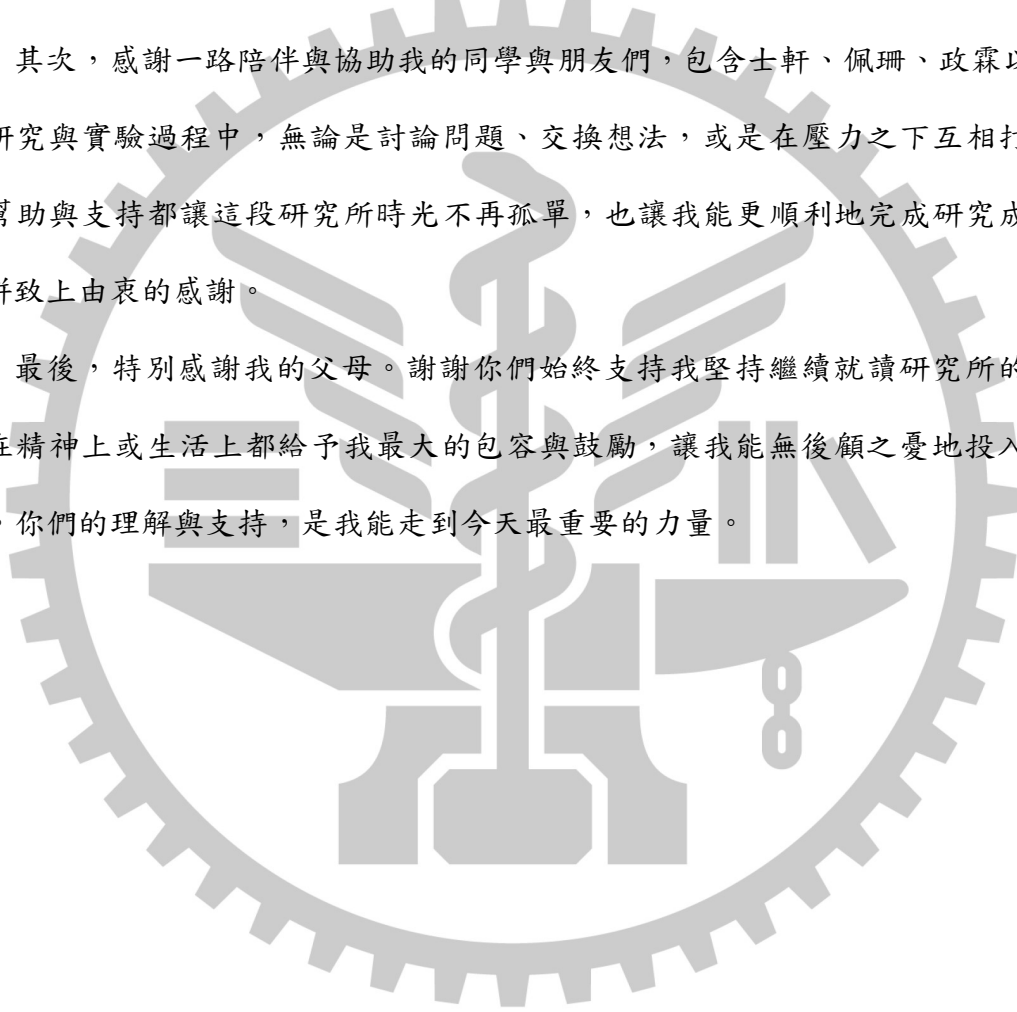
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摘要

去雜訊擴散機率模型 (DDPM) 近年在醫學影像分割任務中表現良好，然而多數現有方法仍需透過額外的條件網路 (如條件嵌入或特徵融合模組) 將參考影像資訊引入擴散過程，這類設計不僅可能造成條件資訊在編碼過程中的壓縮，也使模型架構與推論流程更加複雜。為了滿足臨床皮膚影像分割在準確度與實際部署效率上的需求，本研究提出 Modulated Prior Diffusion (MPD)，以調制擴散模型的先驗分佈的方式取代傳統的顯式條件注入機制。具體而言，MPD 在反向擴散初始化階段，直接以參考影像為中心的高斯分佈進行取樣，使條件資訊在擴散過程一開始即被保留，並自然地影響後續的去雜訊生成，而無需額外的條件網路設計。

在實驗驗證方面，本研究於公開的 ISIC 皮膚病灶資料集以及高雄醫學大學 KMU 臨床臉部影像資料集上進行評估，並以 IoU 與加權 IoU 作為效能指標。實驗結果顯示，所提出之 MPD 在維持精簡模型架構的情況下，於不同資料集上皆能達到穩定且具競爭力的分割表現，顯示該方法具備良好的跨資料集泛化能力。整體而言，MPD 透過更為簡潔的條件設計，在不增加模型複雜度的前提下，達成有效且穩定的醫學影像分割效果，並兼顧實際臨床應用需求。

關鍵字：去雜訊擴散機率模型, 醫學影像分割, 擴散先驗調制, 條件擴散模型, 輕量化
模型設計



Modulated Prior Diffusion for Medical Image Segmentation

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Abstract

Denoising Diffusion Probabilistic Models (DDPMs) have recently shown strong performance in medical image segmentation. However, most existing methods still rely on additional conditioning networks, such as conditional embeddings or feature fusion modules, to introduce reference image information into the diffusion process, which may lead to compression of conditional information and increased model and inference complexity. To address the requirements of clinical skin image segmentation in terms of accuracy and practical deployment efficiency, this study proposes Modulated Prior Diffusion (MPD), which replaces conventional explicit conditioning by modulating the prior distribution of the diffusion model. Specifically, MPD initializes the reverse diffusion process by sampling from a Gaussian distribution centered on the reference image, thereby preserving conditional information from the start of the diffusion process without introducing additional conditioning network designs.

The proposed method is evaluated on the ISIC dataset and a clinical facial image dataset from Kaohsiung Medical University (KMU), using IoU and weighted IoU as evaluation metrics. Results show that MPD achieves stable and competitive segmentation performance across datasets while maintaining a compact architecture. Overall, MPD enables effective medical image segmentation with a simpler design suitable for clinical applications.

Keywords: Denoising Diffusion Probabilistic Models, Medical Image Segmentation, Modulated Diffusion Prior, Conditional Diffusion Models, Lightweight Model Design.



Table of contents

摘要.....	i
Abstract.....	iii
Table of contents	v
List of Figures	vii
List of Tables.....	ix
1 Introduction.....	1
1.1 Background.....	1
1.2 Motivation.....	2
1.3 Challenge and Proposed Approach.....	3
1.4 Contributions.....	4
2 Related Work	5
2.1 Deep Learning-based Medical Image Segmentation	5
2.2 Foundation Model-based Segmentation: MedSAM	6
2.3 Diffusion-based Segmentation: SegDiff	7
3 Proposed method.....	8
3.1 Key Formulation of Modulated Prior Diffusion	8
3.2 Architectural Comparison between GPCD and MPD	10
3.3 Condition Strength and Forward Process Modification.....	12
3.3.1 Condition Strength Parameter w	12

3.3.2	MPD Forward Process Modification	14
3.4	Training and Sampling Procedure of MPD.....	16
3.4.1	Training Procedure.....	16
3.4.2	Sampling Procedure	17
3.5	Theoretical Analysis of Modulated Prior Diffusion	18
3.5.1	Loss Function Comparison	19
3.5.2	Information-Theoretic Interpretation.....	20
4	Experiments and Results.....	24
4.1	Datasets	24
4.2	Experimental Setup.....	27
4.3	Evaluation Metrics	30
4.4	Experimental Results	31
4.4.1	Model Size Comparison.....	31
4.4.2	Effect of the Condition Strength Parameter w	32
4.4.3	Comparison with Baseline Methods	38
5	Conclusions.....	41
	References.....	42

List of Figures

1.1	Conceptual illustration of the general medical image segmentation pipeline. Different types of medical images are processed by a segmentation model to generate corresponding segmentation masks.	2
2.1	Overview of the MedSAM architecture. The framework consists of a pretrained image encoder and a mask decoder for medical image segmentation.	6
2.2	Overview of the SegDiff architecture. Conditional features are extracted by an image encoder and fused with noisy mask features at each diffusion step to progressively refine the segmentation result.	7
3.1	Initialization of the reverse diffusion process in Gaussian Prior Conditional Dif- fusion (GPCD). The reverse process starts from a pure Gaussian prior, and the initial noisy sample does not contain conditioning information from the refer- ence image. Conditioning is introduced only through network modules during the subsequent denoising process.	9
3.2	Initialization of the reverse diffusion process in Modulated Prior Diffusion (MPD). The reverse process starts from a modulated prior whose mean is shifted to- ward the reference image. As a result, the initial noisy sample already contains structural information related to the reference image, allowing conditioning to influence the diffusion process from initialization.	9

3.3	Architectural comparison between (a) Gaussian Prior Conditional Diffusion (GPCD) and (b) Modulated Prior Diffusion (MPD). The two frameworks differ in how conditioning information is incorporated into the generative process. . .	12
3.4	Conceptual illustration of the condition strength parameter w . Increasing w increases the proportion of reference information in the modulated prior while reducing the relative contribution of random noise.	14
3.5	Conceptual illustration of the modified forward process in MPD.	15
3.6	One-dimensional conceptual illustration of the modulated prior distribution. . .	15
3.7	Training pipeline of MPD. The noise term is sampled from the modulated prior distribution.	16
3.8	Sampling process of MPD. The reverse process is initialized from the modulated prior distribution.	17
4.1	Example image and ground-truth mask from the ISIC dataset.	25
4.2	Example image and ground-truth mask from the KMU dataset.	27
4.3	Effect of the condition strength parameter w on IoU for the ISIC dataset. . . .	33
4.4	Segmentation results on the ISIC dataset under different values of the condition strength parameter w (from left to right: input image, ground truth, and predicted masks corresponding to different values of w).	35
4.5	Effect of the condition strength parameter w on IoU for the KMU dataset. . . .	36
4.6	Segmentation results on the KMU dataset under different values of the condition strength parameter w (from left to right: input image, ground truth, and predicted masks corresponding to different values of w).	37

List of Tables

4.1	Summary of the ISIC dataset.	25
4.2	Summary of the KMU dataset.	26
4.3	Training configuration of Modulated Prior Diffusion(MPD).	29
4.4	Sampling configuration of Modulated Prior Diffusion (MPD).	29
4.5	Comparison of model complexity in terms of parameter count and model size. .	32
4.6	IoU comparison on the ISIC dataset.	38
4.7	IoU comparison on the KMU dataset.	39
4.8	Segmentation performance on ISIC and KMU datasets with weighted IoU. . . .	40

Chapter 1

Introduction

1.1 Background

Medical image segmentation is a fundamental task in medical image analysis and has been widely applied to various clinical applications, including tumor detection [1], organ delineation [2], and lesion localization [3]. For clinicians, segmentation results often directly influence subsequent diagnosis, treatment planning, and the assessment of disease severity. Therefore, the accuracy and stability of segmentation methods are of critical importance in clinical practice.

With the rapid development of deep learning techniques, numerous deep learning-based segmentation methods have achieved significant progress in medical image analysis [4, 5, 6]. These methods have enabled automated and high-precision medical image segmentation, making reliable and efficient clinical deployment increasingly feasible. A conceptual illustration of the general medical image segmentation pipeline is shown in Figure 1.1.

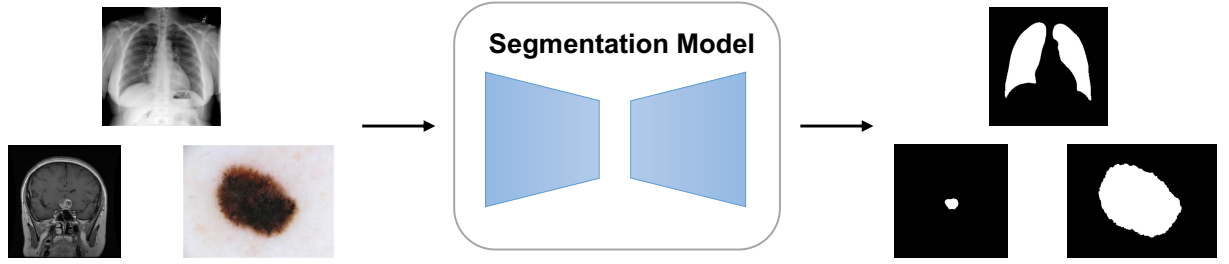


Figure 1.1: Conceptual illustration of the general medical image segmentation pipeline. Different types of medical images are processed by a segmentation model to generate corresponding segmentation masks.

1.2 Motivation

In clinical applications, skin image analysis relies heavily on accurate segmentation of facial skin regions and dermatological lesions. Reliable delineation of these regions is essential for clinical tasks such as disease assessment, treatment planning, and long-term follow-up, especially in scenarios where lesion progression needs to be carefully observed over time.

Recent medical image segmentation methods have demonstrated strong performance. For example, MedSAM learns powerful visual representations through large-scale pretraining [7], while SegDiff [8] applies diffusion models [9] to generate high-quality segmentation results. However, many of these advanced methods are built on large and structurally complex model architectures. Such designs often lead to high computational costs and make deployment in real clinical environments more challenging, as efficiency and resource constraints are important considerations. Therefore, the motivation of this study is to explore whether a more compact and efficient model design can be achieved that reduces model complexity while further improving the accuracy of clinical skin image segmentation.

1.3 Challenge and Proposed Approach

A central challenge in clinical skin image segmentation lies in the fact that high segmentation accuracy is often achieved through large or structurally complex model designs. While such models can deliver strong performance, their complexity increases computational cost and limits their practicality in real clinical environments, where efficiency and resource availability are important factors. As a result, it remains difficult to simultaneously reduce model complexity and further improve segmentation accuracy.

To address this challenge, we propose Modulated Prior Diffusion (MPD), a diffusion-based segmentation framework [9] that incorporates conditional information by directly modulating the diffusion prior. Unlike conventional conditional diffusion approaches that rely on additional conditioning network designs, such as feature fusion networks or conditional embedding mechanisms [8, 10, 11, 12, 13], MPD introduces anatomical guidance at the initialization stage of the reverse diffusion process. By sampling from a Gaussian distribution centered on the reference image, conditional information is preserved from the beginning of the diffusion process without introducing additional conditioning network designs. This design enables a more compact model architecture while maintaining the capability of diffusion models to generate accurate segmentation results.

1.4 Contributions

The main contributions of this work are summarized as follows:

1. A novel conditioning strategy for diffusion-based image segmentation

We propose a novel conditioning strategy for diffusion-based image segmentation, termed Modulated Prior Diffusion (MPD), which incorporates anatomical guidance by modulating the diffusion prior, without additional conditioning networks. This strategy enables the use of anatomical information to guide the segmentation process while keeping the model architecture compact.

2. A condition strength parameter w for random noise and conditional trade-off

We introduce a condition strength parameter w to control the trade-off between random noise and conditional information during the diffusion process, enabling the model to adapt to different segmentation scenarios.

3. Evaluation on public and clinical datasets

We evaluate the proposed MPD framework on the ISIC public benchmark and a clinical facial image dataset from Kaohsiung Medical University (KMU). Experimental results show that MPD achieves segmentation performance comparable to or better than MedSAM and SegDiff while relying on a more compact model design.

Chapter 2

Related Work

2.1 Deep Learning-based Medical Image Segmentation

In recent years, medical image segmentation has attracted significant research attention [4, 5, 6]. Earlier approaches mainly relied on handcrafted features and traditional machine learning techniques, but their performance was often limited when handling complex anatomical structures [14, 15, 16, 17]. With the rise of deep learning, convolutional neural network (CNN)-based methods gradually became the mainstream solution. Architectures such as U-Net and its variants adopt encoder-decoder designs with skip connections, which have substantially improved segmentation accuracy across different medical imaging modalities [4, 6, 18, 19].

After deep learning became dominant, research further moved toward improving model generalization and stability. One direction focuses on foundation model-based segmentation, where large pretrained models are adapted to medical imaging tasks [7]. Another direction explores diffusion-based segmentation, which treats mask prediction as a progressive denoising process [8]. These approaches represent current research trends in medical image segmentation.

Based on these developments, this study selects MedSAM [7] and SegDiff [8] as representative methods of the foundation model-based and diffusion-based paradigms, respectively, and uses them as the background for the proposed Modulated Prior Diffusion (MPD) framework.

2.2 Foundation Model-based Segmentation: MedSAM

MedSAM is a foundation model-based approach for medical image segmentation [7]. It is adapted from the Segment Anything Model (SAM) [20] and leverages large-scale pretraining to learn general visual representations. As shown in Figure 2.1, the framework consists of a pretrained image encoder and a mask decoder. The image encoder extracts high-level features from the input image, and the decoder predicts the corresponding segmentation mask based on these representations.

By inheriting knowledge from large-scale pretraining, MedSAM achieves strong generalization performance across various medical imaging tasks. However, the segmentation performance of MedSAM is closely related to the scale of its backbone network. In general, higher accuracy is achieved by employing larger pretrained encoders. This dependency on model scale leads to increased computational cost and memory consumption, which may restrict its deployment in resource-constrained clinical environments.

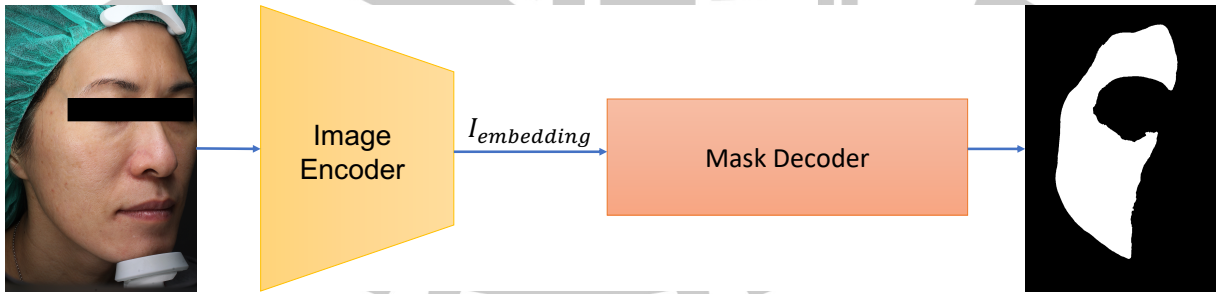


Figure 2.1: Overview of the MedSAM architecture. The framework consists of a pretrained image encoder and a mask decoder for medical image segmentation.

2.3 Diffusion-based Segmentation: SegDiff

SegDiff is a diffusion-based medical image segmentation method that formulates segmentation as a multi-step denoising process [8]. Starting from a noisy mask sampled from a Gaussian distribution, the model progressively refines it through multiple diffusion [9]. As shown in Figure 2.2, conditional features are extracted by an image encoder (e.g., RRDB) and processed through a conditioning module. At each diffusion step, the denoising network fuses the noisy mask features with the conditional features to guide refinement.

However, SegDiff requires an image encoder to extract conditional image features and relies on a conditioning module to introduce these features into the denoising network and fuse them with the noisy mask features, which increases architectural complexity. Moreover, since denoising and feature fusion are repeated across multiple diffusion steps, inference cost is further increased, potentially affecting practical efficiency.

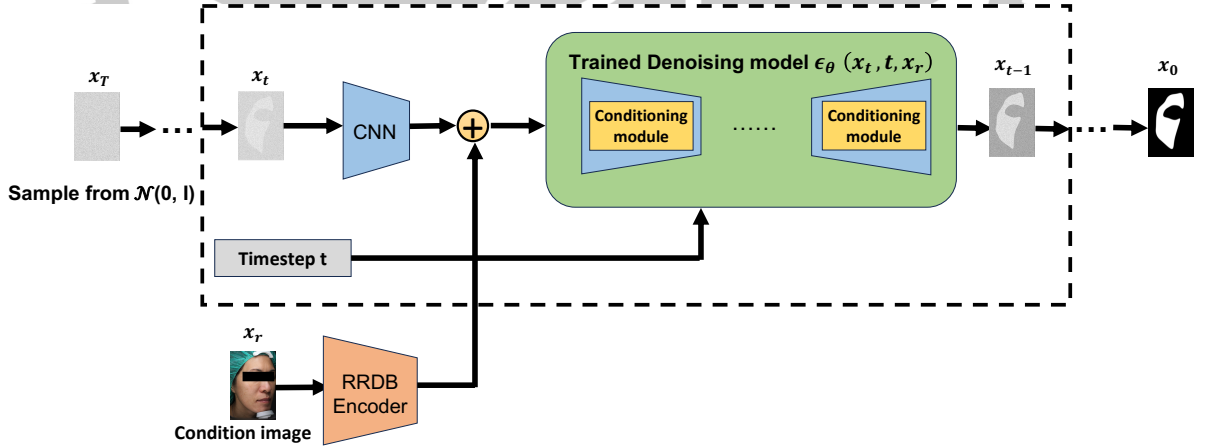


Figure 2.2: Overview of the SegDiff architecture. Conditional features are extracted by an image encoder and fused with noisy mask features at each diffusion step to progressively refine the segmentation result.

Chapter 3

Proposed method

3.1 Key Formulation of Modulated Prior Diffusion

Diffusion-based medical image segmentation commonly adopts a conditional Denoising Diffusion Probabilistic Model (conditional DDPM), in which the reverse denoising process is initialized from a Gaussian prior. We refer to this conventional framework as Gaussian Prior Conditional Diffusion (GPCD) to emphasize that the reverse process starts from a pure Gaussian prior. Specifically, in GPCD, the reverse diffusion process is initialized as:

$$x_T \sim \mathcal{N}(0, I) \quad (3.1)$$

where x_T denotes the initial noisy sample at timestep T . Since the prior mean is zero, the reverse denoising process starts from a random Gaussian noise sample that does not contain conditioning information from the reference image, as illustrated in Figure 3.1. The reference image is incorporated only through conditioning modules inside the denoising network during the iterative process.



Figure 3.1: Initialization of the reverse diffusion process in Gaussian Prior Conditional Diffusion (GPCD). The reverse process starts from a pure Gaussian prior, and the initial noisy sample does not contain conditioning information from the reference image. Conditioning is introduced only through network modules during the subsequent denoising process.

In contrast, the proposed Modulated Prior Diffusion (MPD) initializes the reverse process from a modulated prior whose mean is shifted toward the reference image. Instead of starting from a pure Gaussian prior, MPD initializes the reverse process as:

$$x_T \sim \mathcal{N}(x_r, I) \quad (3.2)$$

where x_T denotes the initial noisy sample at timestep T , and x_r denotes the reference image used as the conditioning input. Compared with GPCD, the reverse denoising process starts from a noisy sample that already contains structural information related to the reference image, as illustrated in Figure 3.2. Consequently, the reverse denoising process is guided from its initialization rather than relying solely on conditioning operations during intermediate denoising steps.



Figure 3.2: Initialization of the reverse diffusion process in Modulated Prior Diffusion (MPD). The reverse process starts from a modulated prior whose mean is shifted toward the reference image. As a result, the initial noisy sample already contains structural information related to the reference image, allowing conditioning to influence the diffusion process from initialization.

3.2 Architectural Comparison between GPCD and MPD

To further clarify the distinction between the two frameworks, we compare Gaussian Prior Conditional Diffusion (GPCD) and Modulated Prior Diffusion (MPD) from the perspective of model design and the manner in which conditioning information is incorporated into the generative process. The architectural difference between the two approaches is illustrated in Figure 3.3.

In GPCD, conditioning information must be introduced into the denoising model to guide the generative process. Specifically, the reference image x_r is first processed by an encoder to obtain a latent feature representation, denoted as $f(x_r)$. At each diffusion step, this feature representation is provided as conditional input to the denoising network and processed through conditioning modules, as illustrated in Figure 3.3(a). Consequently, the generative process does not directly access the original reference image but instead relies on its encoded representation $f(x_r)$.

Since the reference image must first be transformed into a latent feature representation before being utilized by the denoising network, conditioning is performed through this encoded representation. The extent to which structural information is preserved therefore depends on the representational capacity of the encoder and the conditioning modules. In this formulation, the reference information is constrained by the encoding process, which may limit how completely anatomical details are conveyed to the denoising model.

In contrast, MPD adopts a different design strategy. As illustrated in Figure 3.3(b), MPD does not transform the reference image into a separate conditioning representation nor introduce additional conditioning modules, but directly initializes the reverse process from a modulated prior whose mean is shifted toward the reference image, consistent with Equation (3.2) in Sec-

tion 3.1.

Under this formulation, conditioning is realized through prior modulation rather than by encoding the reference image into a feature representation $f(x_r)$ and using it as conditional input to the denoising network. In other words, the reference information is incorporated directly within the prior distribution, instead of being conveyed as an encoded feature representation. This design may conceptually facilitate the preservation of structural information contained in x_r .

Accordingly, the key distinction between the two approaches lies in how conditioning information is introduced into the generative process. GPCD provides conditioning through encoded representations integrated via conditioning modules, whereas MPD integrates conditioning directly into the prior distribution through prior modulation. Compared to GPCD, the prior-modulation design of MPD simplifies the model configuration while potentially enabling more direct utilization of reference information throughout the reverse diffusion process.

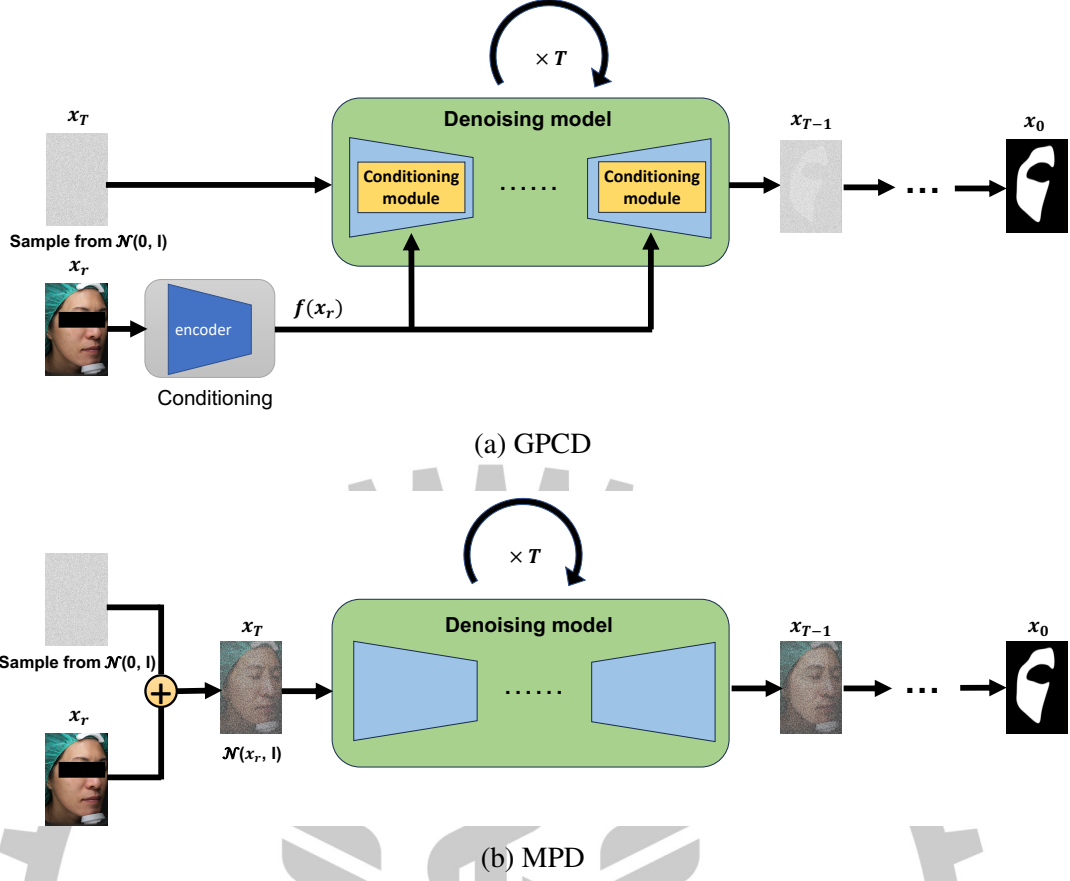


Figure 3.3: Architectural comparison between (a) Gaussian Prior Conditional Diffusion (GPCD) and (b) Modulated Prior Diffusion (MPD). The two frameworks differ in how conditioning information is incorporated into the generative process.

3.3 Condition Strength and Forward Process Modification

3.3.1 Condition Strength Parameter w

Following the formulation in Section 3.1, MPD incorporates the reference image x_r directly into the diffusion model by shifting the mean of the Gaussian prior, so that the reference information becomes part of the diffusion process from initialization. Under this prior-modulation design, we further introduce a condition strength parameter w to control the relative proportion between the reference information and random noise in the modulated prior.

To explicitly describe how this proportion is adjusted, we define the role of the condition

strength parameter from the perspective of the noise term ϵ used in the forward diffusion process. Since the diffusion model is trained by learning to predict the forward noise ϵ , redefining the noise sampling scheme allows the condition strength parameter w to be set consistently during both training and inference, thereby avoiding distribution mismatch.

Specifically, instead of adopting a standard Gaussian prior as in conventional diffusion formulations, MPD employs a modulated prior distribution whose mean and variance are controlled by the condition strength parameter w :

$$\epsilon \sim \mathcal{N}(wx_r, (1-w)I), \quad w \in [0, 1] \quad (3.3)$$

This parameterization provides an interpretable mechanism to regulate conditional guidance strength :

- When w increases, the contribution of the reference image x_r in the modulated prior becomes stronger, while the proportion of random noise decreases. Consequently, the prior distribution is increasingly dominated by the structural information encoded in x_r .
- When w decreases, the proportion of random noise increases and the influence of the reference image x_r is correspondingly reduced.

Overall, the condition strength parameter w provides a controllable mechanism for prior modulation, enabling the model to achieve a balance between structural guidance from the reference image and random noise. For intuitive understanding, the conceptual relationship associated with varying w is illustrated in Figure 3.4.

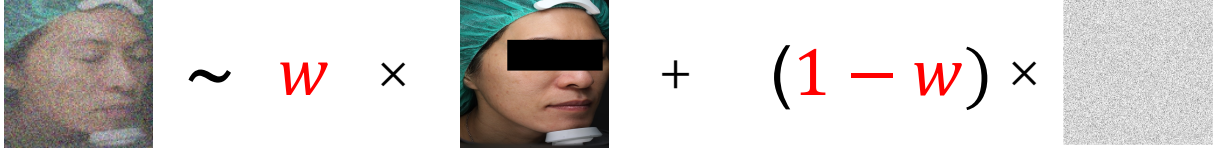


Figure 3.4: Conceptual illustration of the condition strength parameter w . Increasing w increases the proportion of reference information in the modulated prior while reducing the relative contribution of random noise.

Importantly, w is treated as a user-defined parameter that controls the strength of prior modulation. In our experimental design, multiple values of w are evaluated for each dataset, and a corresponding model is trained independently under each setting. The segmentation performance is then compared across different w values to examine how varying the proportion between reference information and random noise influences model behavior.

This design allows us to analyze the effect of prior modulation without altering the model architecture. By adjusting only the condition strength parameter w , which controls the mean and variance of the modulated prior, the strength of conditional information can be regulated without introducing additional conditioning modules or structural modifications.

3.3.2 MPD Forward Process Modification

To maintain consistency with the modulated prior distribution defined in Section 3.3.1, the forward diffusion process is modified accordingly. In Gaussian Prior Conditional Diffusion (GPCD), the forward process is expressed as

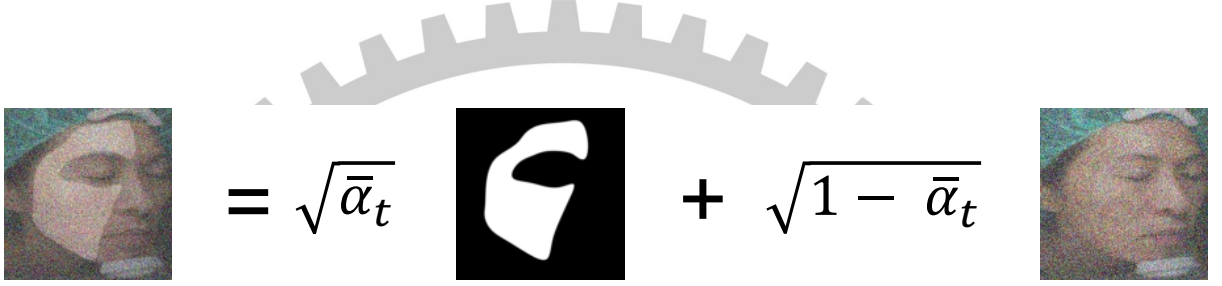
$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (3.4)$$

Under the MPD framework, since the noise term ϵ follows the modulated prior distribution defined in Equation (3.3), the forward process becomes

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(wx_r, (1 - w)I) \quad (3.5)$$

It can be observed that the analytical structure of the diffusion process remains unchanged. The modification lies solely in the sampling distribution of the noise term.

As illustrated in the conceptual illustration shown in Figure 3.5, the noise added to x_0 is sampled from the modulated prior distribution rather than from a Gaussian prior distribution.



$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

Figure 3.5: Conceptual illustration of the modified forward process in MPD.

Figure 3.6 presents a one-dimensional conceptual illustration of the modulated prior distribution. The Gaussian mean shifts from 0 to wx_r , and the distribution becomes narrower as the noise variance decreases.

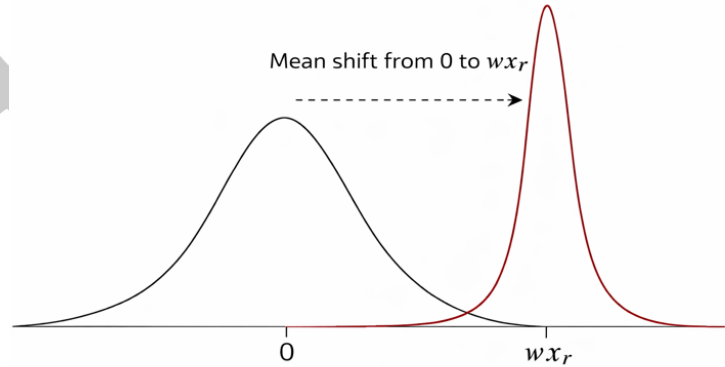


Figure 3.6: One-dimensional conceptual illustration of the modulated prior distribution.

Through this modification, MPD preserves the original formulation of the forward process while embedding conditional information directly into the noise term.

3.4 Training and Sampling Procedure of MPD

This section presents the implementation of Modulated Prior Diffusion (MPD) based on the theoretical formulation introduced in Section 3.3. While the overall framework follows the standard diffusion paradigm, the key difference is that MPD employs a modulated prior during both training and sampling rather than a Gaussian prior.

3.4.1 Training Procedure

The overall training pipeline of MPD is illustrated in Figure 3.7.

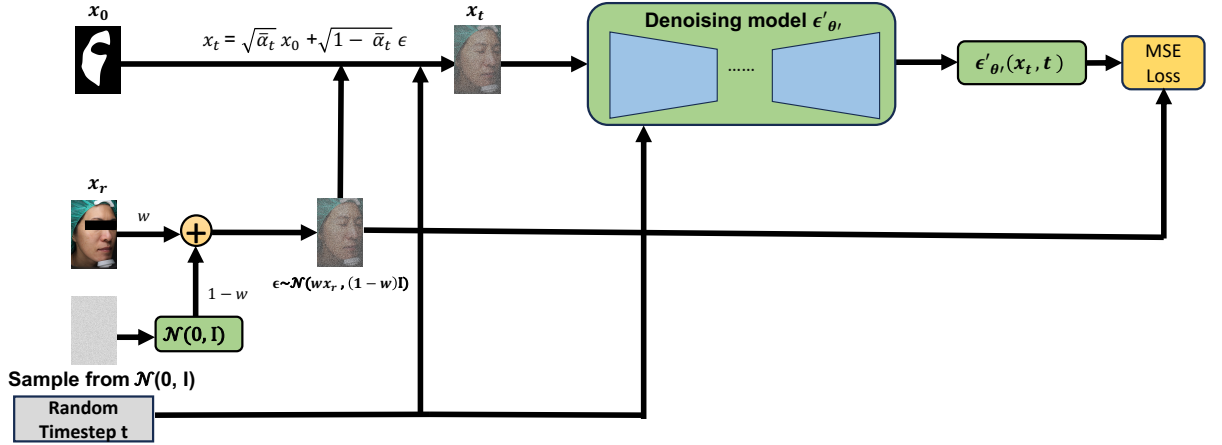


Figure 3.7: Training pipeline of MPD. The noise term is sampled from the modulated prior distribution.

During training, a pair (x_0, x_r) is sampled from the dataset, where x_0 denotes the ground-truth segmentation mask and x_r denotes the reference image. A timestep t is randomly sampled from $\{1, \dots, T\}$. A noise term ϵ is sampled according to the modulated prior distribution defined in Equation (3.3). The noisy sample x_t is then generated by combining x_0 and ϵ through the forward process described in Equation (3.5). The noisy sample x_t and the timestep t are fed into the denoising network to predict the injected noise. The training objective is to minimize the

difference between the predicted noise and the sampled noise. The detailed training procedure is summarized in Algorithm 3.1.

Algorithm 3.1 Training

Input: Ground-truth mask x_0 , reference image x_r , noise schedule $\bar{\alpha}_t$, condition strength parameter w

Output: Trained noise predictor $\epsilon'_{\theta'}$

- 1: **repeat**
 - 2: $(x_0, x_r) \sim p(x_0, x_r)$
 - 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
 - 4: $\epsilon \sim \mathcal{N}(wx_r, (1-w)\mathbf{I})$
 - 5: $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$
 - 6: $\hat{\epsilon} \leftarrow \epsilon'_{\theta'}(x_t, t)$
 - 7: $\mathcal{L} \leftarrow \|\epsilon - \hat{\epsilon}\|^2$
 - 8: Update θ' via backpropagation
 - 9: **until** converged
-

3.4.2 Sampling Procedure

The overall sampling process of MPD is illustrated in Figure 3.8. After training, segmentation masks are generated through the reverse diffusion process.

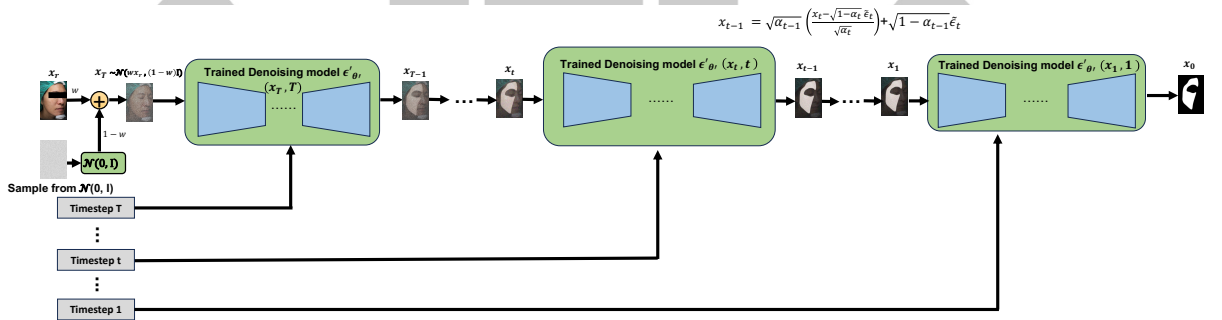


Figure 3.8: Sampling process of MPD. The reverse process is initialized from the modulated prior distribution.

Unlike standard DDPM, which initializes the reverse process from a Gaussian prior distribution, MPD initializes the reverse process by sampling x_T from the same modulated prior

distribution used during training. Subsequently, at each timestep, the current sample x_t and timestep t are fed into the denoising network to predict the noise component. The sample is then updated according to the standard reverse diffusion formulation, progressively removing noise until the final segmentation mask x_0 is obtained. The detailed sampling procedure is summarized in Algorithm 3.2.

Algorithm 3.2 Sampling

Require: x_r : reference image

Require: w : condition strength parameter

- 1: $x_T \sim \mathcal{N}(wx_r, (1-w)\mathbf{I})$
 - 2: **for** $t = T, \dots, 1$ **do**
 - 3: $\tilde{\epsilon}_t = \epsilon'_{\theta'}(x_t, t)$
 - 4: $x_{t-1} = \sqrt{\alpha_{t-1}} \left(\frac{x_t - \sqrt{1-\alpha_t}\tilde{\epsilon}_t}{\sqrt{\alpha_t}} \right) + \sqrt{1-\alpha_{t-1}} \cdot \tilde{\epsilon}_t$
 - 5: **end for**
 - 6: **return** x_0
-

3.5 Theoretical Analysis of Modulated Prior Diffusion

To further analyze the difference between Modulated Prior Diffusion (MPD) and Gaussian Prior Conditional Diffusion (GPCD), this section provides a theoretical comparison from the perspective of the training objective.

Although both MPD and GPCD adopt the same mean squared error (MSE) objective for noise prediction, they differ structurally in how conditioning information is incorporated into the diffusion process.

We examine whether this structural difference may affect the expected denoising error achievable under the same model capacity and training data. Based on this comparison, we further formalize the hypothesis regarding the expected training loss.

3.5.1 Loss Function Comparison

Following the theoretical motivation introduced in Section 3.5, we compare the training objectives of Modulated Prior Diffusion (MPD) and Gaussian Prior Conditional Diffusion (GPCD).

In GPCD, the denoising network aims to predict the noise ϵ added to a ground-truth segmentation mask x_0 at diffusion timestep $t \sim \text{Uniform}(\{1, \dots, T\})$, where ϵ follows a Gaussian prior distribution.

The reference image x_r is first transformed into a conditional feature representation $f(x_r)$, which is then provided to the denoising model together with the noisy sample x_t and the corresponding timestep t . The expected training loss is defined as follows:

$$\mathcal{L}_{GPCD} = \mathbb{E}_{x_0, x_r, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t, f(x_r), t)\|^2] \quad (3.6)$$

In MPD, the same forward diffusion formulation as GPCD is adopted, but the noise ϵ is sampled from a different prior distribution.

The objective of the denoising network is to predict the noise ϵ added to the ground-truth segmentation mask x_0 at diffusion timestep $t \sim \text{Uniform}(\{1, \dots, T\})$, where ϵ follows a modulated prior distribution.

Unlike GPCD, which transforms the reference image x_r into a feature representation $f(x_r)$ and provides it as an additional input, MPD introduces conditional information by directly modulating the prior distribution. Since the noise ϵ is sampled from the modulated prior distribution, the resulting noisy sample x_t inherently contains conditional information from the reference image x_r . Therefore, at each timestep, the denoising network only receives the current noisy sample x_t and the timestep t as inputs. The expected training loss is defined as follows:

$$\mathcal{L}_{MPD} = \mathbb{E}_{x_0, x_r, \epsilon, t} \left[\|\epsilon - \epsilon'_{\theta'}(x_t, t)\|^2 \right] \quad (3.7)$$

Although the two loss functions in (3.6) and (3.7) are analytically similar, the structural difference in how conditioning information is incorporated may influence the effective information available to the denoising network. This distinction does not alter the mathematical form of the optimization objective itself, but it may affect the achievable expected denoising error under the same model capacity and training data.

3.5.2 Information-Theoretic Interpretation

In Section 3.5.1, we discussed that although MPD and GPCD share a similar training loss formulation, their structural designs are different. This structural difference may influence how conditioning information from the reference image is preserved and utilized by the denoising network.

Based on this observation, we formulate the following hypothesis regarding their achievable training performance. To provide theoretical support for this hypothesis, we analyze the two frameworks from an information-theoretic [21]. In particular, we examine how the conditioning pathway in each model affects the relationship between the reference image and the effective input received by the denoising network.

This analysis provides an intuitive explanation of why MPD may achieve a lower expected training loss under the same model capacity and training data.

Hypothesis

Let x_r encode anatomical structure information. Under the same model capacity and training data, we hypothesize that Modulated Prior Diffusion (MPD) is expected to achieve a lower

expected training loss than Gaussian Prior Conditional Diffusion (GPCD) in segmentation tasks, due to its ability to preserve more complete conditioning information in the forward process.

Therefore, under the same model capacity and training data, the expected training loss of MPD is hypothesized to be less than or equal to that of GPCD, which can be expressed as the following inequality:

$$\mathbb{E} \left[\left\| \epsilon - \epsilon'_{\theta'}{}^{MPD}(x_t, t) \right\|^2 \right] \leq \mathbb{E} \left[\left\| \epsilon - \epsilon_{\theta}{}^{GPCD}(x_t, f(x_r), t) \right\|^2 \right] \quad (3.8)$$

Conditioning Path Analysis

To support the above hypothesis, we analyze the conditioning pathway in MPD and GPCD under finite model capacity.

In GPCD, the reference image x_r must pass through a learned transformation $f(x_r)$. Under finite model capacity, this learned transformation may involve compression and potential information loss. Therefore, the conditioning information accessible to the denoising network depends on the amount of information retained in $f(x_r)$.

In contrast, MPD modulates the prior distribution and injects x_r directly and linearly into the generative prior. Consequently, the structural information from the reference image is incorporated into the diffusion process from the initialization stage. In principle, this design ensures full access to anatomical structure within the model input.

According to the Data Processing Inequality (DPI)[22], applying a deterministic transformation to x_r can only decrease or leave unchanged the mutual information between the model input and the original reference image. Therefore, the effective mutual information satisfies:

$$I(Z; x_r)_{MPD} \geq I(Z; f(x_r))_{GPCD} \quad (3.9)$$

where Z denotes the effective input received by the denoising network. This inequality indicates that MPD preserves higher conditional information under the same model capacity and training data.

Information–Risk Perspective

From the perspective of the Information–Risk Tradeoff theorem [21, 23], higher mutual information between the model input and the reference image is associated with a lower achievable risk bound.

As shown in Equation(3.9), MPD preserves higher conditional mutual information than GPCD under finite model capacity. Consequently, the achievable risk bound satisfies:

$$R^*(Z)_{MPD} \leq R^*(Z)_{GPCD} \quad (3.10)$$

where $R^*(Z)$ denotes the achievable risk lower bound under finite model capacity.

As a result, the learned denoising function in MPD is expected to approximate the true noise ϵ with lower variance, yielding a smaller expected training loss under the same model capacity and training data.

This theoretical implication provides an information-theoretic justification for the inequality stated in Equation(3.8). In particular, the lower achievable risk bound of MPD implies that its expected training loss is smaller than or equal to that of GPCD. This suggests that MPD has the potential to achieve improved segmentation performance under the same model capacity and training data.

Although the above analysis provides an information-theoretic rationale for the proposed hypothesis by linking conditioning information preservation to a lower achievable risk bound, the practical validity of this hypothesis ultimately depends on empirical performance. There-

fore, the actual effectiveness of MPD in comparison with GPCD will be further evaluated through comprehensive experimental studies presented in the following section.



Chapter 4

Experiments and Results

This chapter presents the experimental evaluation of the proposed Modulated Prior Diffusion (MPD) framework. The target task is medical image segmentation.

To assess the effectiveness of MPD, we compare it with two representative baseline models, SegDiff and MedSAM. These methods represent diffusion-based and large-scale pretrained foundation-model approaches, respectively, enabling a comparison across different model categories.

Experiments are conducted on two datasets. The KMU Dataset serves as the clinical application dataset, while the ISIC dataset is adopted as a public benchmark to evaluate generalization performance.

Segmentation accuracy is evaluated using Intersection over Union (IoU) as the primary metric. In addition, a weighted IoU calculated according to the number of test samples in each dataset is employed to provide an overall performance measure across datasets.

The detailed experimental setup and results are described in the following sections.

4.1 Datasets

We evaluate the proposed method on two datasets: the publicly available ISIC skin lesion dataset and a clinical facial image dataset collected at Kaohsiung Medical University (KMU).

The two datasets differ in imaging modality, resolution, and application context, allowing us to assess both generalization performance and clinical applicability.

ISIC Dataset

The ISIC dataset is a widely used benchmark for skin lesion segmentation. It consists of dermoscopic images accompanied by expert-annotated ground-truth segmentation masks. The lesion boundaries are often irregular and exhibit low contrast with surrounding skin, which makes accurate segmentation challenging.

The dataset contains 1,279 images with a typical resolution of approximately 1020×760 pixels. A summary of the ISIC dataset is provided in Table 4.1. Representative dermoscopic images and their corresponding ground-truth masks are shown in Figure 4.1.

The dermoscopic imaging modality provides magnified views of skin lesions, highlighting fine-grained texture patterns and boundary details, which contribute to the difficulty of the segmentation task.

Number of images	1279
Image modality	Dermoscopic images
Resolution	$\sim 1020 \times 760$ pixels

Table 4.1: Summary of the ISIC dataset.

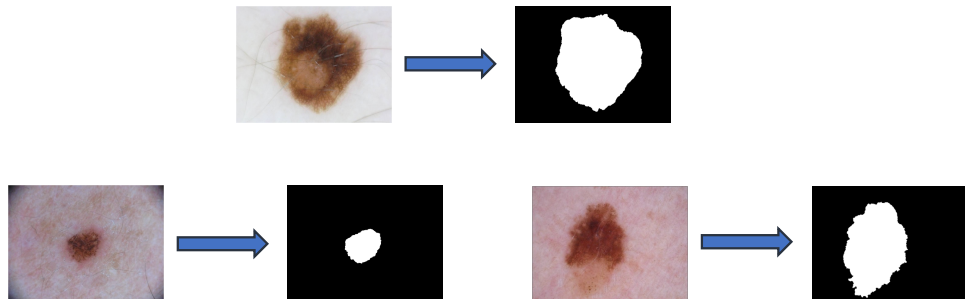


Figure 4.1: Example image and ground-truth mask from the ISIC dataset.

KMU Dataset

The KMU dataset is a clinical image dataset collected at Kaohsiung Medical University for facial skin region segmentation. It consists of high-resolution facial photographs captured in real clinical settings, and all ground-truth segmentation masks were manually annotated by medical professionals.

The dataset contains 249 images with an approximate resolution of 3500×5200 pixels. The images were captured from multiple viewing angles, including left, right, and frontal views, and cover subjects aged between 20 and 80 years.

The main characteristics of the KMU dataset are summarized in Table 4.2. Representative facial images and their corresponding segmentation masks are illustrated in Figure 4.2.

This dataset serves as the primary clinical application dataset for evaluating the practical relevance of the proposed framework.

Number of images	249
Image modality	Clinical facial images
View angles	Left, right, frontal
Resolution	$\sim 3500 \times 5200$ pixels
Age range	20–80 years

Table 4.2: Summary of the KMU dataset.

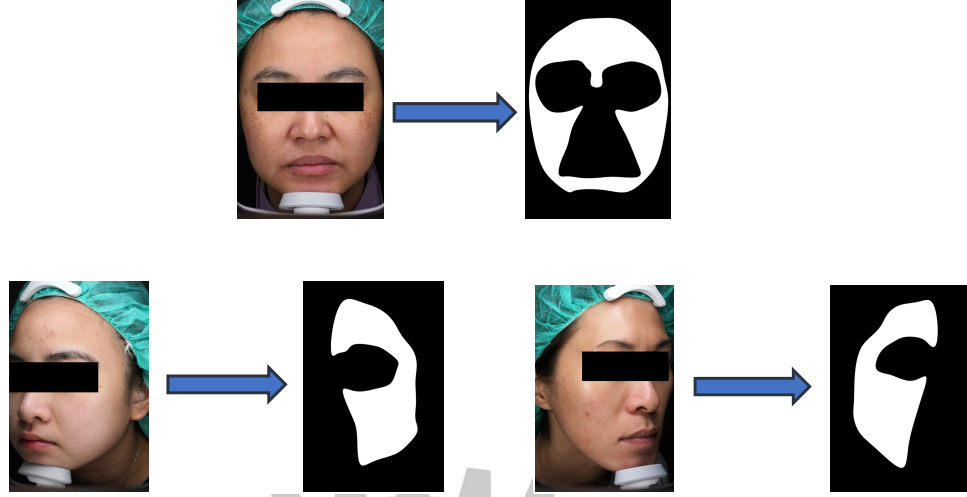


Figure 4.2: Example image and ground-truth mask from the KMU dataset.

The two datasets differ substantially in size. The ISIC dataset serves as a large public benchmark, whereas the KMU dataset is comparatively smaller and clinically collected. If the overall results of the two datasets were directly averaged with equal weight, the dataset with fewer samples could have an excessive influence on the final evaluation, thereby affecting the fairness of the comparison.

To ensure fairness, performance is reported separately for each dataset. In addition, an overall metric weighted by the number of test samples in each dataset is computed to reflect their relative sizes. Based on the relative proportions of the two datasets, the weighting ratio between ISIC and KMU is approximately 8:1. The formal definitions and calculation details of the evaluation metrics are provided later in this section.

4.2 Experimental Setup

To evaluate the proposed Modulated Prior Diffusion (MPD), we compare it with two representative segmentation models: SegDiff [8] and MedSAM [7].

SegDiff is selected because it represents diffusion-based segmentation models. Since our method is also built on a diffusion framework, comparison with SegDiff allows us to examine

whether the proposed modulated prior design provides practical improvements over the conventional Gaussian Prior Conditional Diffusion (GPCD) framework.

MedSAM represents large-scale pretrained foundation models that are widely used in medical image segmentation. By comparing with MedSAM, we aim to examine whether MPD can achieve competitive performance without relying on large pretrained backbones.

For SegDiff and MedSAM, we follow the official implementation and recommended training settings provided by the original authors.

For MPD, the training configuration is designed with reference to commonly adopted hyperparameter settings in Stable Diffusion, with slight adjustments made for the medical image segmentation task. The model is trained for 500 epochs with an initial learning rate of 5×10^{-6} . During the first 50 epochs, a warm-up scheduler is applied to gradually increase the learning rate. After the warm-up stage, a cosine learning rate scheduler is used to progressively reduce the learning rate, which helps improve training stability. We use the AdamW optimizer with a weight decay of 10^{-4} . The diffusion step number T is set to 1000, and the noise schedule is linear. The condition strength parameter $w \in [0, 1]$ is used in both training and sampling stages. In our experiments, w is varied from 0.0 to 1.0 with an interval of 0.1, and a separate model is trained for each value of w to ensure consistency between training and inference.

The denoising network is implemented using a U-Net architecture. We set the number of residual blocks to 2 in each downsampling and upsampling block. The base channel number is 128, and the channel multiplier is [1, 2, 2, 2].

All training images are resized to 128×128 with RGB channels. The model is implemented in PyTorch, and all experiments were conducted on one NVIDIA Tesla V100 GPU. The detailed hyperparameter settings are summarized in Table 4.3.

MPD Training Configuration	
Epochs	500
Batch size	4
Learning rate	5×10^{-6}
Optimizer	AdamW
Weight decay	1×10^{-4}
Diffusion steps (T)	1000
Noise schedule	linear
Condition strength (w)	0.0 – 1.0 (interval 0.1)
Base channels	128
Number of residual blocks	2
Channel multiplier	1, 2, 2, 2

Table 4.3: Training configuration of Modulated Prior Diffusion(MPD).

During inference, DDIM sampling [24] is adopted to accelerate the reverse diffusion process. For each trained model, the condition strength parameter w used during training is kept the same during inference to ensure consistency between training and generation. The sampling hyperparameters are summarized in Table 4.4.

MPD Sampling Configuration	
Diffusion steps (T)	100
Batch size	8
Condition strength (w)	0.0 – 1.0 (interval 0.1)

Table 4.4: Sampling configuration of Modulated Prior Diffusion (MPD).

4.3 Evaluation Metrics

To quantitatively evaluate segmentation performance, we adopt Intersection over Union (IoU) as the primary evaluation metric. IoU is widely used in medical image segmentation as it directly measures the region-level overlap between the predicted mask and the corresponding ground-truth mask, thereby reflecting boundary alignment and segmentation accuracy. Formally, IoU is defined as:

$$IoU = \frac{|P \cap G|}{|P \cup G|} \quad (4.1)$$

where P denotes the predicted segmentation region and G denotes the ground-truth region. The numerator represents the overlapping area between prediction and ground truth, while the denominator represents the area of their union. IoU ranges from 0 to 1, with higher values indicating better segmentation accuracy.

Since our experiments are conducted on two datasets with substantially different sample sizes, namely ISIC and KMU, directly averaging IoU scores across datasets with equal weighting may introduce evaluation bias. Specifically, under equal weighting, the smaller dataset would be assigned the same importance as the larger dataset, thereby disproportionately amplifying its influence on the overall metric. As a result, the final evaluation may not accurately reflect the true performance distribution across the combined sample population, potentially reducing the statistical representativeness and fairness of the aggregated performance measure.

To address this issue, we further compute a dataset-size weighted IoU as an overall cross-dataset evaluation metric. The weighted IoU is defined as:

$$\text{Weighted IoU} = \frac{N_{\text{ISIC}} \cdot IoU_{\text{ISIC}} + N_{\text{KMU}} \cdot IoU_{\text{KMU}}}{N_{\text{ISIC}} + N_{\text{KMU}}} \quad (4.2)$$

where N_{ISIC} and N_{KMU} denote the numbers of test samples in the ISIC and KMU datasets, respectively, and IoU_{ISIC} and IoU_{KMU} represent the corresponding average IoU scores. This formulation ensures that each dataset contributes proportionally according to its sample size, resulting in a more statistically representative and fair overall evaluation.

4.4 Experimental Results

In this section, we present the experimental results of the proposed MPD framework. We first compare model complexity in terms of parameter count and model size. Then, we analyze the effect of the condition strength parameter w on segmentation performance. Finally, we compare MPD with baseline methods on both datasets and report the overall weighted performance.

4.4.1 Model Size Comparison

To evaluate model complexity, we first compare the number of parameters and the model size of MPD with those of the baseline methods. Model complexity is an important factor in practical deployment, as it directly affects memory consumption and computational requirements.

Table 4.5 summarizes the comparison among MedSAM, SegDiff, and the proposed MPD framework. MedSAM contains 90.6 million parameters with a model size of 345.7 MB, while SegDiff has 40.7 million parameters and a size of 155.4 MB. In contrast, MPD requires only 38.5 million parameters and has a model size of 147.0 MB.

Method	Parameters (M)	Model Size (MB)
MedSAM	90.6	345.7
SegDiff	40.7	155.4
MPD (ours)	38.5	147.0

Table 4.5: Comparison of model complexity in terms of parameter count and model size.

These results indicate that MPD maintains a more compact architecture than both baseline methods. Despite having fewer parameters and a smaller model size, MPD is designed to achieve competitive segmentation performance, which will be further analyzed in the following sections.

4.4.2 Effect of the Condition Strength Parameter w

To systematically investigate the role of the condition strength parameter w , we conduct a comprehensive analysis by varying w from 0.0 to 1.0 with an interval of 0.1. For each value of w , an independent model is trained to ensure consistency between training and inference, thereby preventing potential distribution mismatch.

Analyzing the influence of w helps reveal how different levels of conditional strength affect the balance between structural guidance and stochastic noise, and how this balance influences segmentation performance. Through this analysis, we aim to examine whether increasing conditional guidance consistently improves performance and whether an optimal balance exists between structural preservation and randomness.

The following experiments are conducted on both the ISIC and KMU datasets to compare the impact of the condition strength parameter w on segmentation performance under different data characteristics.

Quantitative Analysis on the ISIC Dataset

As shown in Figure 4.3, on the ISIC dataset, the IoU score increases steadily as the condition strength parameter w increases from 0.0 to 0.9. Specifically, the performance improves significantly from 0.303 at $w = 0.0$ to 0.827 at $w = 0.9$. This indicates that increasing w provides stronger guidance through the modulated prior and leads to better segmentation performance. The performance reaches its peak at $w = 0.9$. However, when w is further increased to 1.0, a slight decline to 0.810 is observed, indicating that overly strong guidance does not lead to further improvement.

Notably, at $w = 0.9$, MPD also outperforms both MedSAM and SegDiff. This result suggests that MPD achieves better performance when structural information and stochastic noise are properly balanced via the modulated prior.

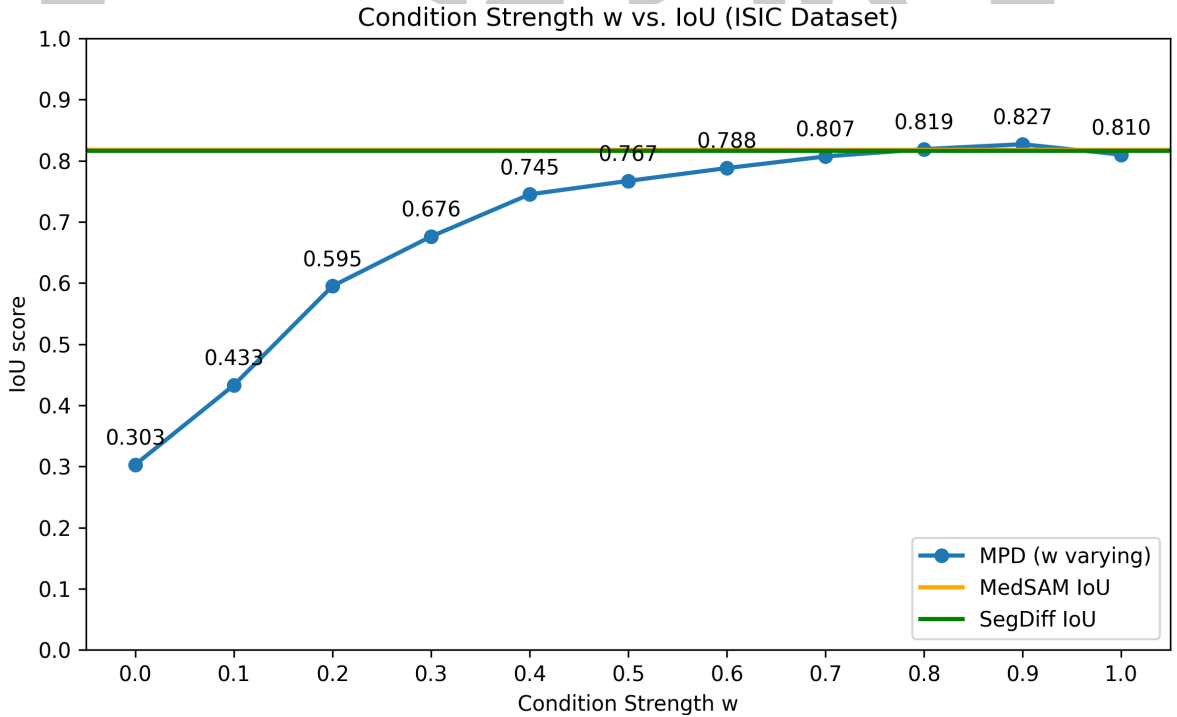


Figure 4.3: Effect of the condition strength parameter w on IoU for the ISIC dataset.

Qualitative Analysis on the ISIC Dataset.

As shown in Figure 4.4, the qualitative results are consistent with the quantitative trend. When $w = 0.0$, the predicted masks are unstable, and some regions are either incomplete or incorrectly segmented. This indicates that when the conditioning strength in the modulated prior is too weak, the model struggles to capture the lesion structure accurately.

As w increases to moderate values (e.g., $w = 0.3$ to $w = 0.6$), the segmentation results become more stable, and the predicted boundaries gradually align with the ground truth. When w reaches higher values such as $w = 0.8$ and $w = 0.9$, the predicted masks are visually closest to the ground truth, with clearer shapes and more consistent boundaries.

When w is further increased to $w = 1.0$, the segmentation results remain similar to those at $w = 0.9$, and no obvious additional improvement can be observed. Overall, increasing w enhances the influence of reference structural information in the modulated prior, leading to more stable and accurate segmentation results within an appropriate range.

Input	Ground truth	W = 0.0	W = 0.1	W = 0.2	W = 0.3	W = 0.4	W = 0.5	W = 0.6	W = 0.7	W = 0.8	W = 0.9	W = 1.0
												
												
												

Figure 4.4: Segmentation results on the ISIC dataset under different values of the condition strength parameter w (from left to right: input image, ground truth, and predicted masks corresponding to different values of w).

Quantitative Analysis on the KMU Dataset

As shown in Figure 4.5, a similar increasing trend is observed on the KMU dataset. As the condition strength parameter w increases, the IoU score improves steadily and reaches its highest value at $w = 0.8$. Specifically, the performance increases from 0.475 at $w = 0.0$ to 0.881 at $w = 0.8$.

When w exceeds 0.8, a slight decrease in IoU can be observed, indicating that further increasing w does not yield additional improvement. Compared to the ISIC dataset, where the best performance is achieved at $w = 0.9$, the KMU dataset reaches its optimal point at a slightly lower value. This difference may be related to variations in data characteristics and structural complexity between the two datasets.

At $w = 0.8$, MPD achieves performance comparable to MedSAM and outperforms SegDiff, demonstrating the effectiveness of the proposed modulated prior mechanism on the KMU dataset.

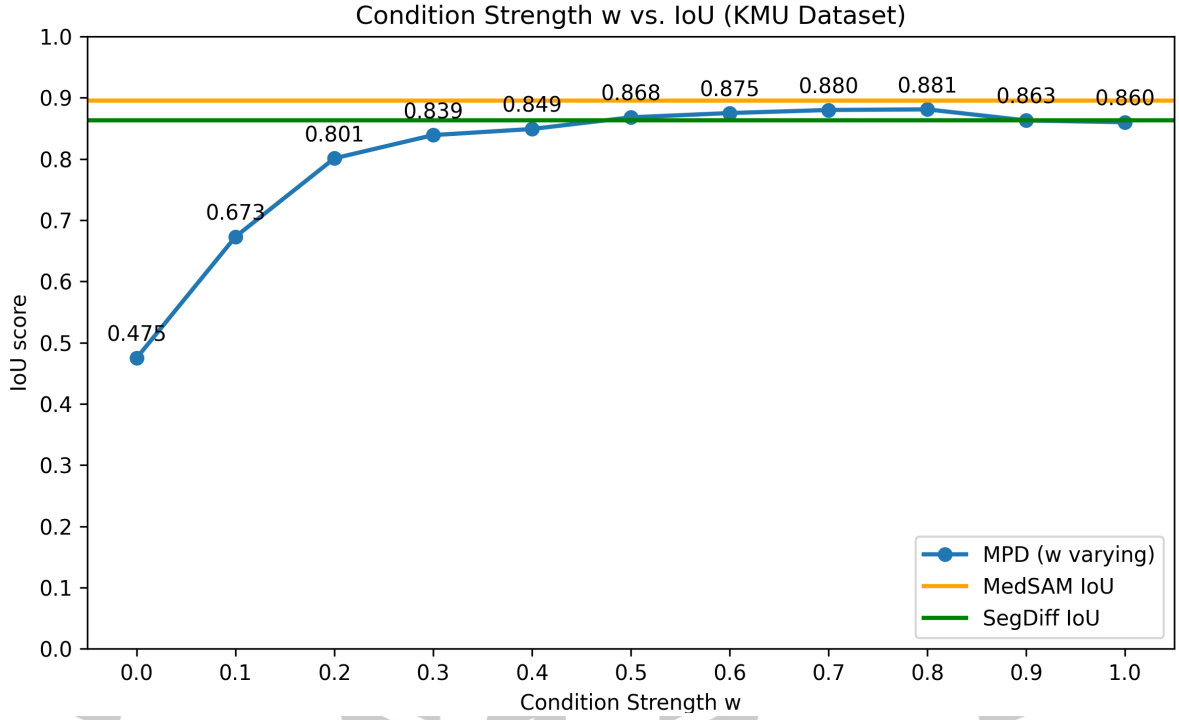


Figure 4.5: Effect of the condition strength parameter w on IoU for the KMU dataset.

Qualitative Analysis on the KMU Dataset

As shown in Figure 4.6, the qualitative results on the KMU dataset are consistent with the quantitative trend. When $w = 0.0$, the predicted masks are incomplete and fail to accurately represent the facial region, indicating unstable segmentation when the conditioning strength in the modulated prior is insufficient.

As w increases (e.g., from $w = 0.3$ to $w = 0.6$), the predicted regions become more coherent and gradually align with the ground truth. When w reaches 0.7 or 0.8, the segmentation results visually resemble the ground truth with clear structural boundaries and consistent region shapes.

When w increases beyond 0.8, only minor visual differences can be observed, and the

overall structure remains largely unchanged. Overall, when w exceeds approximately 0.6, the segmentation results become stable and remain consistent within the range close to the optimal value.

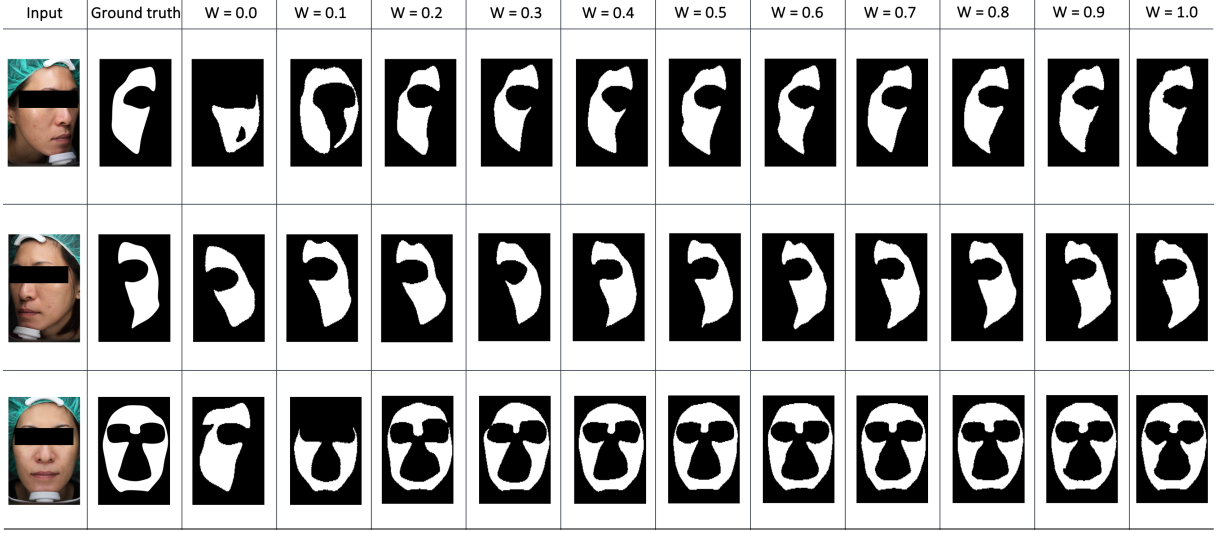


Figure 4.6: Segmentation results on the KMU dataset under different values of the condition strength parameter w (from left to right: input image, ground truth, and predicted masks corresponding to different values of w).

Overall, the experimental results on both the ISIC and KMU datasets demonstrate that the condition strength parameter w plays an important role in the segmentation performance of MPD. As w increases, the IoU score improves, indicating that stronger conditioning helps the predicted masks better align with the ground truth.

However, increasing w does not continuously improve performance. For both datasets, segmentation results stabilize within a certain range, and excessively large values do not provide additional benefit. Moreover, the optimal value of w differs slightly between the two datasets, suggesting that the appropriate conditioning strength may depend on data characteristics.

These findings indicate that properly selecting the condition strength parameter w is essential for achieving stable and accurate segmentation performance.

4.4.3 Comparison with Baseline Methods

Following the analysis of the condition strength parameter w in the previous subsection, we further compare the proposed MPD framework with SegDiff and MedSAM to comprehensively evaluate its segmentation performance.

For the comparison, the condition strength parameter w of MPD is set to the optimal value identified in the previous subsection. Specifically, for each dataset, the w value that achieves the highest IoU performance is selected to reflect the segmentation capability of MPD under its optimal configuration.

Performance on the ISIC Dataset

Table 4.6 summarizes the IoU results on the ISIC dataset. MPD achieves the highest IoU score of 0.827, outperforming MedSAM (0.818) and SegDiff (0.816).

This result indicates that MPD effectively preserves structural guidance through the modulated prior and achieves superior segmentation accuracy without relying on large pretrained backbones or additional conditioning modules.

Method	IoU
MedSAM	0.818
SegDiff	0.816
MPD	0.827

Table 4.6: IoU comparison on the ISIC dataset.

Performance on the KMU Dataset

Table 4.7 presents the IoU results on the KMU dataset. MedSAM achieves the highest IoU score of 0.895, while MPD achieves 0.881, outperforming SegDiff (0.863).

Although MPD does not obtain the absolute highest score on this dataset, it demonstrates competitive performance while maintaining a more compact architecture.

Method	IoU
MedSAM	0.895
SegDiff	0.863
MPD	0.881

Table 4.7: IoU comparison on the KMU dataset.

Cross-Dataset Overall Performance

Since the ISIC and KMU datasets differ substantially in size, we further compute a dataset-size weighted IoU as a cross-dataset evaluation metric. Table 4.8 summarizes the segmentation performance of all methods on the ISIC and KMU datasets, together with the dataset-size weighted IoU.

MPD achieves the highest weighted IoU score of 0.833, outperforming MedSAM (0.827) and SegDiff (0.822). This indicates that, after accounting for dataset size differences, MPD demonstrates the best overall segmentation performance.

Overall, MPD achieves superior performance on the public benchmark dataset, competitive performance on the clinical dataset, and the best weighted performance across datasets.

Method	ISIC IoU	KMU IoU	Weighted IoU
MedSAM	0.818	0.895	0.827
SegDiff	0.816	0.863	0.822
MPD	0.827	0.881	0.833

Table 4.8: Segmentation performance on ISIC and KMU datasets with weighted IoU.



Chapter 5

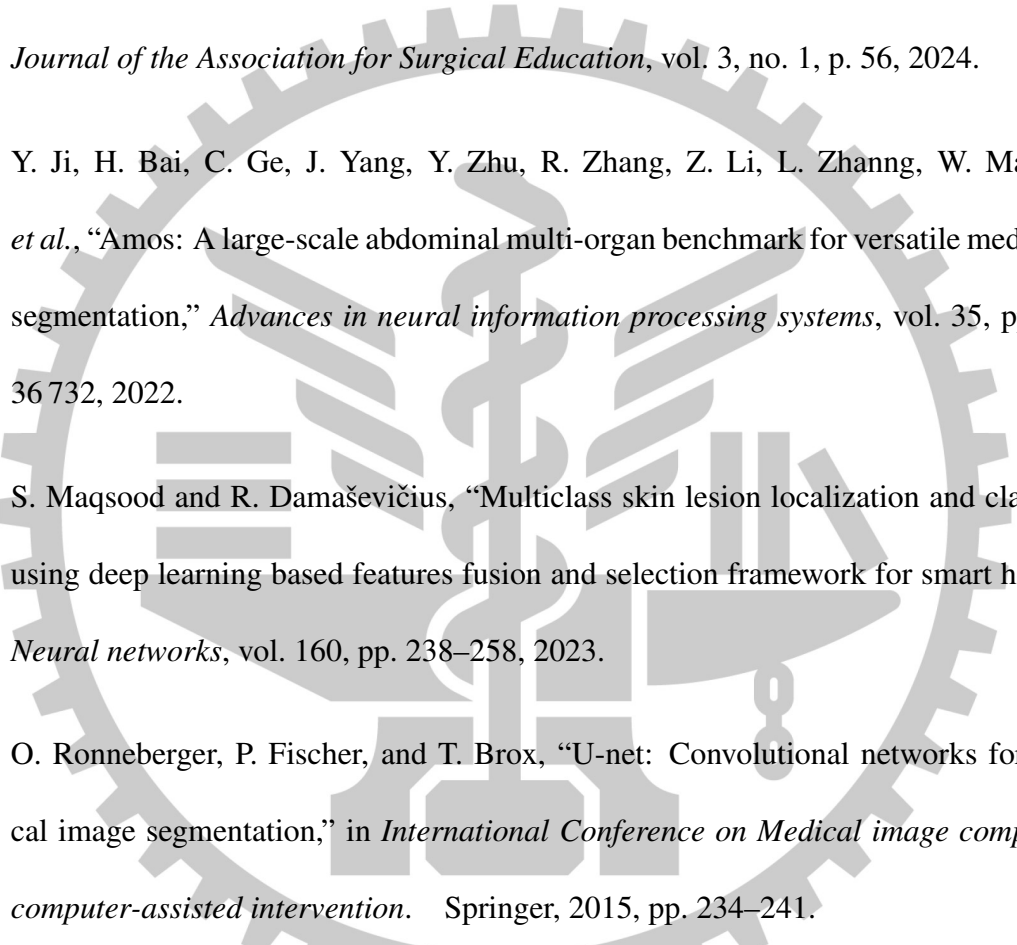
Conclusions

This study presents Modulated Prior Diffusion (MPD), a conditional diffusion framework designed for medical image segmentation. Unlike approaches that introduce additional encoders or complex feature fusion mechanisms to incorporate conditioning information, MPD directly employs a modulated prior distribution at the initial stage of the reverse diffusion process. By embedding reference information into the prior distribution itself, the proposed design naturally preserves structural guidance throughout the diffusion process.

Experimental results demonstrate that MPD achieves comparable or superior IoU performance on both the KMU clinical dataset and the ISIC public benchmark when compared with MedSAM and SegDiff. While maintaining competitive segmentation accuracy, MPD retains a compact and efficient architecture, reducing reliance on large pretrained backbones or heavy network designs.

Furthermore, the proposed condition strength parameter w provides controllable conditional guidance within the diffusion process. As w increases, segmentation performance improves until reaching a saturation point, highlighting the importance of balancing structural guidance and stochastic noise. Overall, MPD offers a simple yet effective framework for conditional diffusion-based medical image segmentation by integrating conditioning information directly into the diffusion prior.

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